

BAWA AMARNATH SANIA

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EDUCATION

National University of Singapore, Singapore

Aug 2024 – Jun 2028

Bachelor of Computing (Honours) – Computer Science, GPA – 4.12/5

Relevant Courses: Software Engineering, Data Structures & Algorithms, Intro to AI & ML, OOP, Intro to OS, Computer Organisation

Victoria Junior College, Singapore

Jan 2022 – Dec 2023

GCE A Levels – 87.5 RP

SKILLS

Technical Skills: MongoDB, PostgreSQL, ReactJS, Java, Go, Python, R, Spring Boot, Git, AWS EC2, ML, MS Excel

Languages: English, Tamil

PROJECTS

AlphaReturns – XRPL Fintech Transaction Platform – Fintech Hackathon 2026

Jan 2026

- Built a full-stack fintech application enabling secure XRP Testnet transfers through a web dashboard and Xaman mobile wallet signing. Refactored legacy modules using object-oriented design principles, improving code scalability and maintainability.
- Integrated XRPL, Xaman SDK, and asynchronous transaction workflows, handling transaction creation, mobile signing, validation, and real-time status updates.
- Implemented polling and WebSocket listeners to handle asynchronous signing outcomes including successful, rejected, and expired transactions.
- Applied knowledge of blockchain transaction lifecycles, account reserves, transaction fees, and secure wallet interaction without exposing private keys.

Project: BudgetBuddy – Smart Budget Tracking Web Application

May 2025 – July 2025

- Developed and deployed a full-stack budgeting platform using React, Spring Boot, MongoDB, and AWS for shared expense tracking, debt simplification, and currency conversion.
- Built a FastAPI ML microservice to analyse spending patterns and generate personalised financial insights with real-time analytics.
- Delivered a feature-complete application awarded Apollo 11 (Advanced) by integrating ML insights, data visualisation, and scalable backend services.

WORK EXPERIENCE

Application Security Intern, Singapore Airlines, Singapore

June 2026 – Present

- Gained exposure to Identity Governance & Administration (IGA), Privileged Access Management (PAM), and application management systems, including their workflows, access controls, and operational processes
- Supported application security assessments and vulnerability testing using Burp Suite to identify and investigate potential security issues.
- Explored threat modelling methodologies and the application of LLMs and AI to support security analysis and improve assessment workflows
- Supported automation initiatives aimed at streamlining operational processes and improving efficiency and reducing manual effort

CCA

Vice Project Director, Freshmen Social Camp, School Of Computing, NUS

Dec 2024 - Jun 2025

- Co-led planning and execution of a 4-day orientation camp for 250+ freshmen, improving first-year engagement and integration.
- Managed a 100-member committee and coordinated 4 subcommittees, ensuring smooth operations, safety, and risk management.